

DURING A RECENT interaction with students of an engineering institute, Navin Bishnoi, Country Head of fabless chip design firm Marvell India, was surprised at how keen the students were to know more about new-age subjects like AI and ML. But barely anyone was interested in the chips, or semiconductors that power them. Only when Bishnoi started going deep into AI applications and its underlying infrastructure that students start to realise the importance of semiconductors. “I realised that the exposure to semiconductors is lesser than other technologies. Engineering students are choosing the easier path to employability compared to semiconductor-hardware engineering, which requires a master’s degree and years of experience.”

The chips Bishnoi discussed are high-end ones. But as semiconductors are used in almost everything electronic around us, their demand has gone through the roof in the past few years. The global chips market, pegged at \$600 billion in 2021, is also expected to grow to \$1 trillion by 2030, per McKinsey. But experts say that current manufacturing capacity is just not sufficient to meet this booming demand.

Consequently, as new fabs—where chips are printed, and then taken through the ATMP (assembly, testing, marking, and packaging) process—are set up around the world and in India, even the demand for professionals qualified in electronics engineering and related

fields is going to surge exponentially. And while the job profiles may sound too technical, most of the learning and upskilling in this industry happens while on the job.

TALENT CRUNCH

Currently, nearly 277,000 people across the US work in the chip-making industry—in R&D, design, manufacturing, testing, etc. The country’s Semiconductor Industry Association (SIA) expects the new fabs that are coming up there to create 42,000 new permanent jobs by 2027, with an average of 185,000 temporary jobs (related to building fabs) being created annually from 2021 to 2026. And this is just the case in the US. Be it the EU, Japan, China, South Korea, or Taiwan, every government is aiming to bring chip production onshore. India, too, has embarked on a journey to develop its own silicon ecosystem. The country—never one of the chip-making nations—is now witnessing interest from large business houses, such as Vedanta and Tata. That gives credence to the burgeoning requirement and opportunity for engineering talent in the country.

More so because semiconductor talent is in short supply. With more fabs coming up, the situation is expected to get worse by the end of 2023, and the rest of the decade. Deloitte estimates that the global semiconductor workforce—pegged at more than 2 million direct employees in 2021—will need to grow by more than 1 million by 2030, adding more than 100,000 workers annually. It is this exponential demand for talent that India can cater to with its crop of engineering students that join the workforce every year. After all, close to 2.5 million students enrol for undergraduate engineering degrees in India annually, with close to 600,000 students opting for the electronics stream, per Statista.

“There will be a strong demand on recruiting processing, equipment, and IC testing engineers, and capacity planning managers,” says Helen Chiang, Head of Asia/Pacific Semiconductor Research and General Manager of IDC Taiwan. She adds that people with knowledge of, or who major in electronics, mechanical, chemical, and materials engineering will be the most sought after. Already, semiconductor job openings in India have increased by 7 per cent between March 2019 and 2023, data from jobs portal Indeed shows.

JOB PROSPECTS

The chip-making industry is a resource-intensive one that requires highly-skilled workers at every step of the process, right from chip design and manufacturing to

THE JOBS AND SALARIES ON OFFER FOR BEGINNERS

Nature	R&D	ATMP	Foundry
► Job profile	► Work on various aspects of chip design and technology development	► Conduct IC testing and packaging, understand related process technologies, evaluation and verification of wafer packaging materials, etc.	► Conduct and manage the whole process of manufacturing wafers/chips, improve process flow, evaluate and manage risks/issues, conduct detection and monitoring analysis and implement new processes
Beginner's salary (in India, not MNCs)	₹3 lakh to ₹18 lakh per annum	₹7 lakh to ₹14 lakh per annum	₹10 lakh to ₹20 lakh per annum initially to entry-level professionals

SOURCE: IDC, AICTE, BT RESEARCH